

KAREEM HEGAZY

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EXPERIENCE

Summary: ML research scientist (Stanford Physics PhD): transformers, neural operators, physics foundation models; 1,330+ citations (*Science*, *Nature Chemistry*, *NeurIPS*, *ICLR*, *AISTATS*); Managed a 12-person team.

UC Berkeley / ICSI — Postdoctoral Researcher, Scientific Machine Learning Aug 2023 - Present

- **NeurDE** (Submitted to JCP): Co-developed a physics-ML hybrid framework (Boltzmann-BGK kinetic theory) for long-horizon forecasting of nonlinear conservation laws.
 - Multi-node distributed training and inference on 80 A100 GPUs; accurately forecasts very challenging systems: supersonic shocks, plasma, turbulence ($Re = 100,000$).
 - Extremely stable: trained on $O(100)$ – $O(1000)$ time steps, accurately forecasts $O(1000)$ steps.
 - Leverages thermodynamic principles to stabilize NeurDE at extremely coarse resolutions and extract subgrid information; extending to hypersonic shocks and porous-media flow.
- **Physics Foundation Model** (first-author paper in preparation): Leading a 12-person team (masters, PhDs, postdocs) scaling NeurDE into a physics foundation model; curating an $O(100)$ TB training dataset.
 - Built highly accurate physics simulators and a randomized data-generation scheme yielding unlimited training data, overcoming the bottlenecks of expensive simulation and scarce existing data.
- **Powerformer** (*AISTATS 2026*): Designed a transformer with weighted causal attention encoding recency bias and local structure; achieved state-of-the-art results on standard time-series forecasting benchmarks.
- **Zero-Shot Super-Resolution** (*ICLR 2026*): Demonstrated fundamental limitations of zero-shot super-resolution in machine-learned operators on PDEs, challenging a widely held assumption in the field.
- **Popcornn** (submitting to *Nature Computational Science*): Co-developed a framework parameterizing reaction paths as continuous neural networks to find chemical transition states on ML interatomic potentials.
 - First double-ended method benchmarked at high-throughput scale ($O(1000)$ reactions); consistently outperforms decades-old gold standards (NEB, GSM) with fixed default settings.
 - Built **Padaquad**, a GPU-parallelized adaptive numerical path integrator with orders-of-magnitude speedup.

SLAC — PhD Researcher, Ultrafast Molecular Dynamics

Jun 2014 - Aug 2023

- **Bayesian Inference for Geometry Retrieval** (*Nature Communications Physics*): Independently derived and implemented a novel Bayesian method reconstructing molecular structures from diffraction data.
 - Improved 3D molecular imaging resolution by $100\times$ while removing the simulation bottleneck.

Google X — AI Resident, Neural Network Fundamentals

Jun 2019 - Dec 2019

- **Adversarial Example Transferability:** Probed fundamental properties of learned representations: built a large ($\sim 10M$) diverse adversarial dataset and measured cross-architecture transferability.

CERN — Researcher, High Energy Physics

Sep 2010 - Sep 2014

- **Higgs Boson Measurement:** Primary contributor to the Higgs spin measurement.

EDUCATION

Stanford University: Physics PhD — Advisers: Phil Bucksbaum, Ryan Coffee

Sep 2014 - Aug 2023

- Thesis: Merging ultrafast gas-phase diffraction experiment, theory, and machine learning for molecular dynamics
- Machine Learning: CS229/ML, CS221/AI, CS231n/CNN, CS224n/NLP

University of Michigan: B.Sc in Physics and Math with Highest Honors

Sep 2009 - Jun 2014

- Awards: William L. Williams Award (best thesis), Angell Scholar (consecutive straight A's), Tiffany Fellowship

SKILLS

- **ML Methods:** Transformers, neural operators, foundation models, Bayesian inference, adversarial robustness
- **Languages & Frameworks:** Python, C/C++, PyTorch, JAX, TensorFlow, $\text{\LaTeX}$
- **Infrastructure:** Multi-node distributed GPU training (80 A100s), GPU-parallelized integration, ML pipelines

SELECTED PUBLICATIONS

1. K. Hegazy, E. Yuan et al., “Neural Network Path Optimization for Finding Transition States on a Machine Learned Potential.” Submitting to *Nature Computational Science*, 2026.
2. M. Sakarvadia, K. Hegazy et al., “The False Promise of Zero-Shot Super-Resolution in Machine-Learned Operators.” *ICLR*, 2026.
3. K. Hegazy, M.W. Mahoney, N.B. Erichson, “Powerformer: A Transformer with Weighted Causal Attention for Time-Series Forecasting.” *AISTATS*, 2026.
4. J. Benitez, K. Hegazy et al., “Neural Equilibria for Long-Term Prediction of Nonlinear Conservation Laws.” Under review at *NeurIPS*, 2026.
5. K. Hegazy et al., “Entropically-Stabilized Computational Grids for Fast and Stable PDE Modeling.” Under review at *NeurIPS*, 2026.
6. K. Hegazy et al., “TorchPathDiffeq: Finding Your Path.” In preparation, 2026.
7. K. Hegazy, V. Makhija et al., “Applying Bayesian Inference and Deterministic Anisotropy to Retrieve the Molecular Structure from Gas-Phase Diffraction Experiments.” *Communications Physics* (Nature), 6, 325, 2023.
8. K. Hegazy et al., “Tracking Dissociation Pathways of Nitrobenzene via MeV Ultrafast Electron Diffraction.” *J. Phys. B*, 57, 195101, 2024.
9. J. Yang, X. Zhu, T.J.A. Wolf, ... K. Hegazy et al., “Imaging CF3I Conical Intersection and Photodissociation Dynamics with Ultrafast Electron Diffraction.” *Science*, 361, 64–67, 2018.
10. E.G. Champenois, D.M. Sanchez, ... K. Hegazy et al., “Conformer-Specific Photochemistry Imaged in Real Space and Time.” *Science*, 374, 178–182, 2021.
11. T.J.A. Wolf, D.M. Sanchez, ... K. Hegazy et al., “The Photochemical Ring-Opening of 1,3-Cyclohexadiene Imaged by Ultrafast Electron Diffraction.” *Nature Chemistry*, 11, 504–509, 2019.
12. Y. Liu et al., “Rehybridization Dynamics into the Pericyclic Minimum of an Electrocyclic Reaction Imaged in Real-Time.” *Nature Communications*, 14, 2795, 2023.
13. Y. Liu et al., “Spectroscopic and Structural Probing of Excited-State Molecular Dynamics.” *Physical Review X*, 10, 021016, 2020.
14. E.G. Champenois et al., “Femtosecond Electronic and Hydrogen Structural Dynamics in Ammonia.” *Physical Review Letters*, 131, 143001, 2023.
15. J. Yang et al., “Diffractive Imaging of Coherent Nuclear Motion in Isolated Molecules.” *Physical Review Letters*, 117, 153002, 2016.
16. S. Li et al., “Characterizing Isolated Attosecond Pulses with Angular Streaking.” *Optics Express*, 26, 4531–4547, 2018.
17. A. Wadell, ... K. Hegazy et al., “Foundation Models for Discovery and Exploration in Chemical Space.” *arXiv:2510.18900*, 2025.